A reproducible computational framework for decoupled wildfire ignition and propagation benchmarking in the Gran Chaco (2018–2025)

D2 targets, M3/M2 inference, and an open benchmark against Canadian FWI and INPE

Abstract

Wildfire machine learning often suffers from spatial leakage and the limited transferability of boreal fire-weather indices to subtropical biomes. For Paraguay's 1-km² Chaco–Oriental grid (2018–2025), we introduce D2, a computational target operator that formally separates primary ignition from tactical propagation, and a gated inference and evaluation framework that makes target construction, leakage control and model comparison explicit. D2 defines ignition as fire on day t with no fire within 25 km on day $t-1$ and no same-cell fire in the previous 10 days; propagation is evaluated within 25 km of a day $t-1$ fire. A biophysical gate removes non-combustible and wet cell-days. A hybrid LightGBM model (M3) predicts ignition without day $t-1$ fire inputs, using M2 as a fallback for cells without history and M1 as a biophysical baseline. On 2023 validation, M3 reaches an area under the receiver operating characteristic curve (ROC-AUC) of 0.7844/0.7628 (Chaco/Oriental) versus 0.7817/0.7496 for M2. In 2023 spatial blocks, M2 is most robust (0.6773/0.7470); M1 does not beat M2 on cells without history. An independent confirmation holdout also retained M2 (0.6871/0.6621). On the retrospective 2024–2025 test, M3 reaches 0.8021/0.7534 versus 0.5173/0.5564 for the Canadian Forest Fire Weather Index (FWI) and 0.5389/0.4990 for the INPE Risco de Fogo; propagation reaches 0.9039/0.7859.

Population-level ignition precision is 0.082 %/0.198 %, confirming that the system is a nowcasting and prioritization tool rather than a stand-alone warning system. The contribution lies in the open D2 operator, gated inference and evaluation harness, released as reproducible software.

Keywords: Wildfire nowcasting; GeoAI; reproducibility; spatial cross-validation; LightGBM; Gran Chaco.

1. Introduction

Vegetation fires in the South American Gran Chaco are a recurring threat to ecosystems, regional climate and public health (Baumann et al., 2017; Flannigan et al., 2009; Jolly et al., 2015). The region combines some of the fastest agricultural frontiers in South America with prolonged dry seasons and extreme heat (Alvarado et al., 2018; Pivello et al., 2021; Vallejos et al., 2015). Operational fire danger there is still largely based on fire-weather indices developed for temperate or tropical savanna systems: the Canadian Forest Fire Weather Index (FWI) (San-Miguel-Ayanz et al., 2012; Stocks et al., 1989; Van Wagner, 1987) and the Instituto Nacional de Pesquisas Espaciais (INPE) Risco de Fogo (Setzer et al., 2019). These indices are widely used, but they were not designed for the moisture-limited, fuel-limited and strongly anthropogenic ignition regime of the Dry Chaco.

Machine learning has become a common alternative for fire susceptibility mapping (Barmpoutis et al., 2020; Jain et al., 2020), typically with gradient-boosted trees (Ke et al., 2017). Two methodological problems limit the operational value of many published models. First, when spatial lags or $t-1$ fire variables enter the ignition layer, the model learns persistence rather than ignition; reported discrimination can be high while the model is unable to anticipate new ignitions (Dormann et al., 2007; Legendre, 1993; Ploton et al., 2020; Roberts et al., 2017; Tobler,

1970). Second, fire-weather indices calibrated in boreal or temperate systems do not necessarily transfer to subtropical biomes, and their failure is rarely quantified on a common retrospective test. A reproducible benchmark is therefore needed not only to compare models, but to make the target definition, the leakage controls and the evaluation protocol explicit enough to be audited. The two tasks are also physically distinct: primary ignition is controlled by fine-fuel curing, atmospheric water demand and human land management, whereas tactical propagation is governed by thermal radiation and the proximity of active perimeters. Conflating both processes into a single binary label does not only introduce a statistical shortcut; it also mixes two different physical regimes that respond to different predictors.

The operational context adds a third constraint: most agencies need a daily product that can run on commodity hardware, with transparent inputs and a documented failure mode. A model that cannot be reproduced or audited is difficult to adopt even when its aggregate score is high. The framework is therefore designed around explicit targets, a frozen test and a small, inspectable software stack.

We address these methodological and operational constraints over the continuous 2018–2025 Visible Infrared Imaging Radiometer Suite (VIIRS) record (Suomi-NPP, NOAA-20 and NOAA-21) across a 1-km² national grid of Paraguay (400 006 cells) (Schroeder et al., 2014). While keeping the underlying learner intentionally replaceable, we formalize four reusable computational contributions:

- **D2 target operator.** A formal target contract that separates primary ignition from tactical propagation, prevents persistence from being mixed with ignition and makes leakage checks explicit.

- **Gated M3/M2 inference.** A two-stage policy that combines biophysical screening with a recurrence-aware hybrid model (M3) where fire history exists and a historical-anchor fallback (M2) where it does not.
- **Reproducible evaluation harness.** A unified protocol that integrates temporal holdout, spatial blocks, date-grouped bootstrap, lead-time analysis, conformal coverage and cost-loss evaluation.
- **Open software and data.** A vectorized, zero-copy Polars/DuckDB pipeline with public code, archived DOIs and automated tests.

The novelty is computational and infrastructural rather than algorithmic: D2 formalizes the target contract, gated inference defines the operational policy, and the software exposes that contract to other learners. The geoscientific contribution is a transferable, leakage-controlled benchmark: full FWI and INPE are tested under the same D2 protocol in the Gran Chaco, and the temporal versus spatial limits of the hybrid model are reported explicitly. The national case study is a testbed for data-scarce subtropical regions; we do not claim universal validity.

The remainder of the paper is organized as follows. Section 2 defines the D2 operator, the biophysical gate, the decoupled inference architecture and the evaluation harness. Section 3 reports selection on 2023 validation, the retrospective 2024–2025 test, the benchmark against FWI and INPE, the spatial and temporal generalization analyses and the operational utility. Section 4 discusses why canonical indices fail, what decoupling buys and where it fails, and the limitations of the study. Section 5 concludes.

2. Computational framework

2.1 D2 target operator

The target operator D2 is applied to the national 1-km² grid and to all daily VIIRS observations from 2018 to 2025. For a cell i and day t:

- **Primary ignition** is a fire at (i, t) when there is no fire within 25 km at $t-1$ and no fire in cell i during the previous 10 days.
- **Tactical propagation** is a fire at (i, t) within 25 km of at least one fire at $t-1$.

The two domains are disjoint. This prevents a model from obtaining high discrimination by learning "fire continues next to yesterday's fire" while being evaluated as an ignition model. The same operator is used for training, validation, test and alert generation. The complete sampling counts and spatial stratification are reported in Supplementary Note S1 and Table S10.

The population is sampled at a 1:10 ratio per task and region. Training uses 2018–2022, validation uses 2023 and the test is the frozen 2024–2025 period. The ignition test contains 376 552 cell-days (34 232 positives) and the propagation test contains 1 437 414 cell-days (130 674 positives). The combined task-level test therefore has 1 813 966 cell-days. The sample is case-control; population denominators are kept separately and used for prior correction. The same test rows are used for every model and for FWI/INPE, so the comparison is paired by construction.

Formally, let F_t be the set of cells with a thermal detection on day t. D2 ignition at cell i and day t requires $i \in F_t$, no cell in F_{t-1} within 25 km of i, and no detection in cell i during the ten preceding days. D2 propagation requires $i \in F_t$ and at least one cell in F_{t-1} within 25 km. The 25-km threshold is a domain restriction, not a negative-sampling buffer; no 20-km buffer is applied to the test sample. The buffer is used only in the spatial block holdout described below.

2.2 Biophysical gate and decoupled inference

A first-stage gate removes cell-days with no combustible fuel, with vapor pressure deficit (VPD) below 0.8 kPa, or with 48-h rainfall of at least 10 mm. Gating is applied before feature construction; it reduces the inference space while retaining 97.17 % of thermal detections and removing 21.7 % of cell-days. The gate is not used as a label filter: it only removes physically implausible candidates. Water, consolidated urban surfaces and saturated fuels are therefore excluded from inference, but they remain part of the audit population used to compute retention. Two models share the gated space. M3 is a hybrid LightGBM model with weather, fuel, ecotone, anthropogenic priors and causal historical features. It contains no t-1 fire variable. M2 is a base LightGBM model with weather, fuel, ecotone and a historical anchor. In the operational layer, M3 is used for cells with observed fire history ($freq_hist_1km > 0$) and M2 is used as a fallback for cells without history; M1 remains a biophysical baseline. The propagation layer uses a separate model with a causal kernel at t-1: distance to the nearest active fire, fire counts within 25 km and accumulated fire radiative power (FRP). The architecture is summarized in Figure 1. M1 uses 24 weather and fuel variables, M2 adds the ecotone and a 25-km historical anchor (35 variables), and M3 adds local density, annual-cycle and accessibility terms (44 variables in Chaco and 45 in Oriental). The feature families are deliberately separated: M3 receives no t-1 fire variable, whereas propagation receives only causal t-1 fire information. This separation makes the ablation interpretable and prevents the fallback from becoming a hidden persistence model. M3 groups meteorology (maximum and minimum temperature, thermal range, VPD, wind, 48-h rainfall, consecutive dry days, Fine Fuel Moisture Code (FFMC) and Initial Spread Index (ISI)), fuel fractions (pasture, savanna, dry forest, humid forest and wetland/palm), ecotone, civil calendar and burning season, accessibility corridors, local density at 5 and 10 km,

and causal recurrence. M1 contains only the meteorological and fuel subsets. The feature lists are frozen in the code release. The fallback rule is evaluated on cells with freq_hist_1km = 0; M2 is the validated fallback and M1 is retained only as a diagnostic baseline.

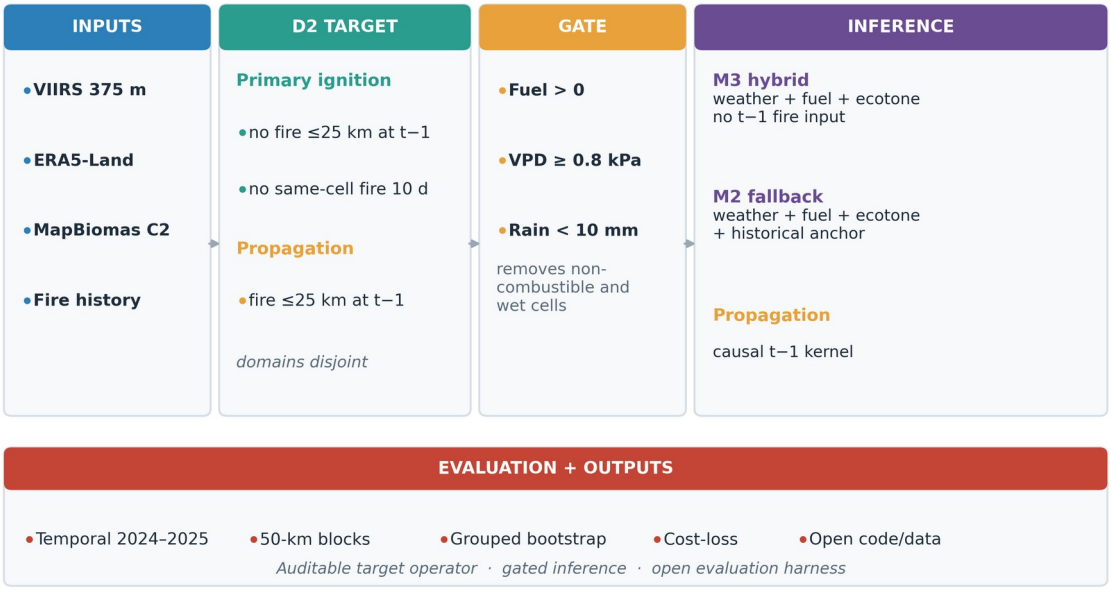


Figure 1: Computational framework: multi-source inputs, D2 target operator, biophysical gate, decoupled M3/M2 ignition and t-1 propagation, reproducible evaluation and open outputs.

Probabilities are calibrated with a monotonic bounded Platt transform. The monotonic constraint preserves the ranking of the tree ensemble while preventing the slope inversion that can occur under severe class imbalance (Niculescu-Mizil & Caruana, 2005; Platt, 1999). The calibration is fitted on 2023 and frozen for the 2024–2025 test. Calibration details, including the monotonicity proof and the full hyperparameter table, are in Supplementary Note S2 and Table S2.

The same calibration and cutoff logic is used for M1, M2, M3 and propagation. Cutoffs are selected on validation with an F0.5 ($\beta = 0.5$) criterion and then frozen. The operational alert applies the cutoff after prior correction, not before; otherwise the population prevalence would change the ranking threshold. This distinction is important because a cutoff that looks conservative on a 1:10 sample can be extremely permissive at population prevalence.

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2.3 Reproducible evaluation harness

The evaluation harness is part of the computational contribution. It contains five components:

1. A strictly temporal split: training 2018–2022, validation 2023, test 2024–2025.
2. A spatial block holdout of 50 km with a 20-km buffer, used to separate temporal from spatial generalization.
3. A paired bootstrap grouped by date ($B = 1\,000$) for confidence intervals and differences against FWI and INPE.
4. A lead-time evaluation using t-1 weather to quantify the nowcast-to-forecast gap.
5. A population-level and cost-loss evaluation to move beyond case-control precision.

The bootstrap resamples dates rather than rows, because daily fire activity is strongly temporally clustered. For each replicate, all cells from the sampled dates are retained; this preserves within-day dependence and gives more honest intervals than row-wise resampling. The same block structure is used for paired differences against FWI and INPE. The spatial holdout divides the country into 50-km blocks, removes an entire block and a 20-km buffer from training, recomputes the historical priors without the held-out fires and evaluates both held-in and held-out cells. The primary metrics are the area under the receiver operating characteristic curve (ROC-AUC) and precision–recall area under the curve (PR-AUC). We also report Brier score, expected calibration error, precision, recall, F0.5 and false positives at the validation cutoff. AUC confidence intervals use 500 grouped bootstrap replicates; paired model differences use 1 000 replicates. Because four main comparisons are made against FWI and INPE, the significance statements should be read as a conservative screening result rather than as multiplicity-adjusted inference. The bootstrap resolution is 0.001, so p-values are reported as ≤ 0.001. The protocol is applied without modifying the test set. The target operator, the list of test

cell-days and the evaluation scripts are archived with the code release. The complete protocol is reported in Supplementary Note S3 and Tables S7–S14.

2.4 Data and implementation

The grid contains 400 006 cells of 1 km² in Universal Transverse Mercator (UTM) zone 21S. Daily inputs are VIIRS 375 m active fire detections from Suomi-NPP, NOAA-20 and NOAA-21 (Schroeder et al., 2014); ERA5-Land 3-hourly meteorology aggregated to daily values (Muñoz-Sabater et al., 2021); annual MapBiomas Collection 2 maps (2017–2023), joined to each row with the prior-year map (MapBiomas Chaco, 2023); Hansen Global Forest Change (GFC-2025 v1.13) tree-cover loss at t-1 and t-2 (Hansen et al., 2013); and causal historical priors built with a monthly cutoff and a rolling three-year window. Raw data are public but are not redistributed; processed data and trained models are archived with a DOI.

The pipeline uses a deterministic integer cell_id, a NumPy lookup in O(1), and columnar joins in Polars and DuckDB (Raasveldt & Mühleisen, 2019). A spatial join of 50 000 points is shown only as a reference (Supplementary Table S4); the operational lookup avoids point-in-polygon at inference. The full inference pipeline processes 400 006 cells in 3.474 s with 2 212.9 MB peak RAM on ARM64; x86_64 was not measured. The study domain and regional split are shown in Figure 2; the per-stage profile is in Supplementary Table S4.

Data quality checks are part of the pipeline: duplicate VIIRS overpasses are collapsed, low confidence detections are removed, daily precipitation is reconstructed from sub-daily increments and impossible values above 300 mm/day are rejected. ERA5-Land nodes are mapped to cells with a precomputed cKDTree. The processed archive contains the test targets, model probabilities, benchmark layers, sampling plan and trained artifacts, so the result tables can be regenerated without downloading the full raw archive. The complete data dictionary and

checks are in Supplementary Note S1 and Table S15. The repository includes 76 unit tests, a pinned dependency file, a continuous-integration workflow and a public GitHub release. The test suite covers target definitions, VIIRS filtering, climate aggregation, monotonic calibration, metrics and the alert layer; it can run without downloading the raw archive. This separation between a reproducible evaluation package and the full raw-data pipeline is deliberate: reviewers can audit the tables without a multi-terabyte download, while the canonical scripts remain available for a full rebuild.

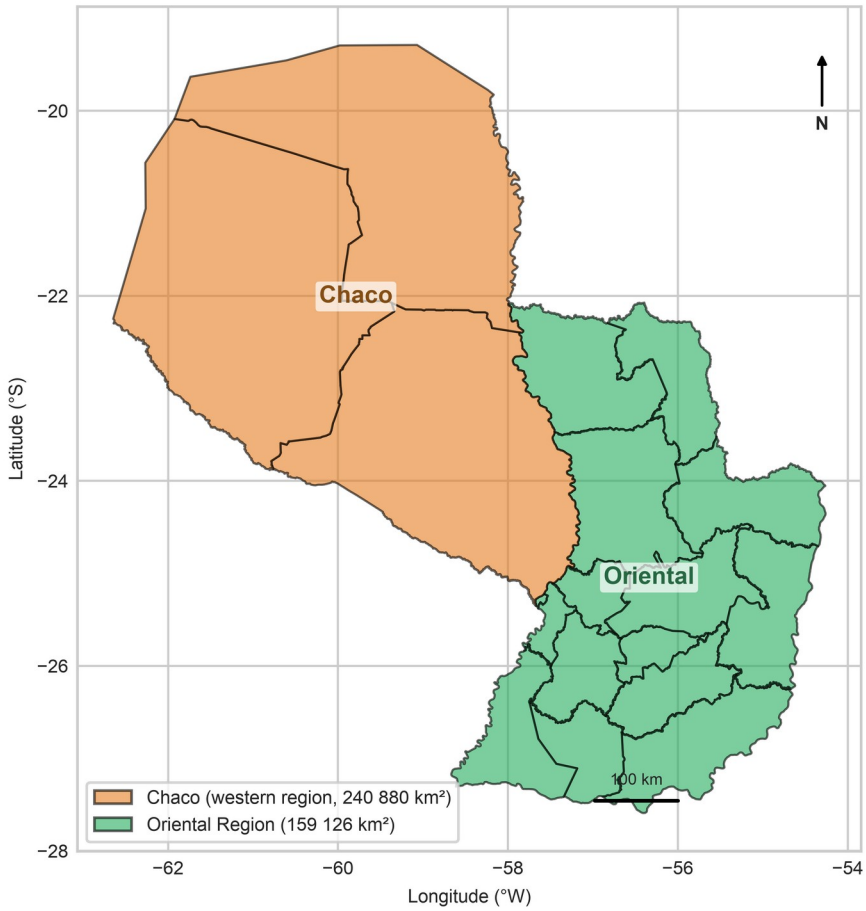


Figure 2: Study domain and regional split: Dry Chaco (west) and Oriental Region (east) over the national 1-km² grid (EPSG:32721).

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3. Results

3.1 Selection on 2023 validation and retrospective test

Model selection used only the 2023 validation set and the 2023 development spatial holdout. The 2024–2025 test was not used to select the model, variables or thresholds. Table 1 summarizes temporal validation.

Table 1: 2023 temporal validation (1:10 case-control). Selection was frozen before the 2024–2025 test was evaluated. The M1 baseline is reported in Supplementary Table S16.

Region	Model	ROC-AUC	PR-AUC	F0.5
Chaco	M2 base	0.7817	0.2452	0.2933
Chaco	M3 hybrid	0.7844	0.2542	0.2979
Oriental	M2 base	0.7496	0.2221	0.2631
Oriental	M3 hybrid	0.7628	0.2441	0.2875

The paired M3 − M2 difference was +0.0028 [+0.0001, +0.0054] ROC-AUC and +0.0089 [+0.0034, +0.0144] PR-AUC in Chaco, and +0.0132 [+0.0097, +0.0169] and +0.0221 [+0.0166, +0.0272] in Oriental (grouped-by-date bootstrap, B = 1 000). With the pre-specified margin δ = 0.01, M3 is retained in both regions. In 2023 spatial blocks, M2 was the most robust model (0.6773/0.7470), ahead of M3 (0.6496/0.7277). M1 was evaluated as a diagnostic baseline and did not outperform M2 on cells without history; M1 is therefore rejected as the operational fallback (Supplementary Tables S8 and S16). The final architecture is M3 for cells with history and M2 for cells without history.

Table 2 reports the retrospective 2024–2025 test. M3 reaches ROC-AUC 0.8021 in Chaco and 0.7534 in Oriental. Propagation reaches 0.9039/0.7859. Case-control F0.5 precision is 32.82 %/30.13 % for ignition and 83.69 %/65.03 % for propagation. These are sample-level metrics; they are not operational precision. The actual gated M3/M2 policy is close to M3 but not uniformly better: in Chaco it slightly improves ROC-AUC while reducing precision; in Oriental it improves precision over M3 but reduces ROC-AUC. The full sample and population comparison is in Supplementary Table S17.

Table 2: Retrospective D2 test (2024–2025; 1 813 966 task cell-days in total: 376 552 ignition and 1 437 414 propagation). Precision and recall are case-control (9.09 % prevalence) at the F0.5 cutoff. The test was already observed and is not used for selection. FP = false positives; CI = confidence interval; F0.5 = F-beta with β = 0.5. Population metrics are in Supplementary Table S6.

Region	Model	ROC-AUC [95 % CI]	PR-AUC	Precision (%)	Recall (%)	FP
Chaco	M3 ignition	0.8021 [0.797–0.807]	0.2842	32.82	30.30	9 960
Chaco	M3/M2 gated	0.8035 [0.799–0.809]	0.2843	32.70	30.87	10 200
Chaco	M2 ignition	0.7991 [0.794–0.805]	0.2766	33.28	27.26	8 773
Chaco	Propagation	0.9039 [0.899–0.909]	0.6935	83.69	52.40	5 353
Oriental	M3 ignition	0.7534 [0.747–0.759]	0.2340	30.13	18.59	7 835
Oriental	M3/M2 gated	0.7502 [0.744–0.756]	0.2327	30.15	18.55	7 811
Oriental	M2 ignition	0.7406 [0.735–0.746]	0.2146	26.09	19.59	10 075
Oriental	Propagation	0.7859 [0.779–0.794]	0.3752	65.03	19.84	8 351

Figure 3 shows the corresponding ROC and precision–recall curves for the hybrid ignition and propagation layers. The figure separates the two tasks and two regions, so the trade-off between the high discrimination of propagation and the much lower discrimination of ignition is visible without a second benchmark layer. The curves are descriptive; the operating points in Table 2 are those used by the alert layer. Ignition ROC curves rise steeply only at relatively high false-positive rates, and the corresponding precision–recall curves remain close to the case-control baseline. Propagation curves are farther from the diagonal in both regions, especially in Chaco, consistent with the stronger event signal in that task. The visual gap between ROC-AUC and PR-AUC is expected at 9.09 % prevalence and reminds us that a high ROC-AUC can coexist with modest precision.

Population-level ignition precision is 0.082 % in Chaco and 0.198 % in Oriental. Propagation precision is 9.67 % and 4.78 %. These are the operationally relevant values and are much lower than the case-control values. Relative to prevalence, ignition precision is about 5.4× and 4.8× above chance, while propagation precision is about 62× and 23× above chance. Absolute precision remains low; the model is therefore a prioritization and nowcast layer, not a stand-alone alert system.

On the retrospective test, M3 does not uniformly dominate M2. In Chaco, M3 improves ROC-AUC and PR-AUC, but M2 has slightly higher precision and fewer false positives; in Oriental,

M3 improves precision and reduces false positives, with lower recall. The gain is a trade-off, not a universal superiority. By year, M3 keeps ROC-AUC 0.8030/0.8011 in Chaco and 0.7532/0.7539 in Oriental for 2024/2025; propagation falls from 0.9069 to 0.8982 in Chaco and from 0.7911 to 0.7785 in Oriental. 2024 uses the 2023 fuel map with lag 1 and 2025 uses 2023 with lag 2; metrics are therefore reported separately rather than as a single average.

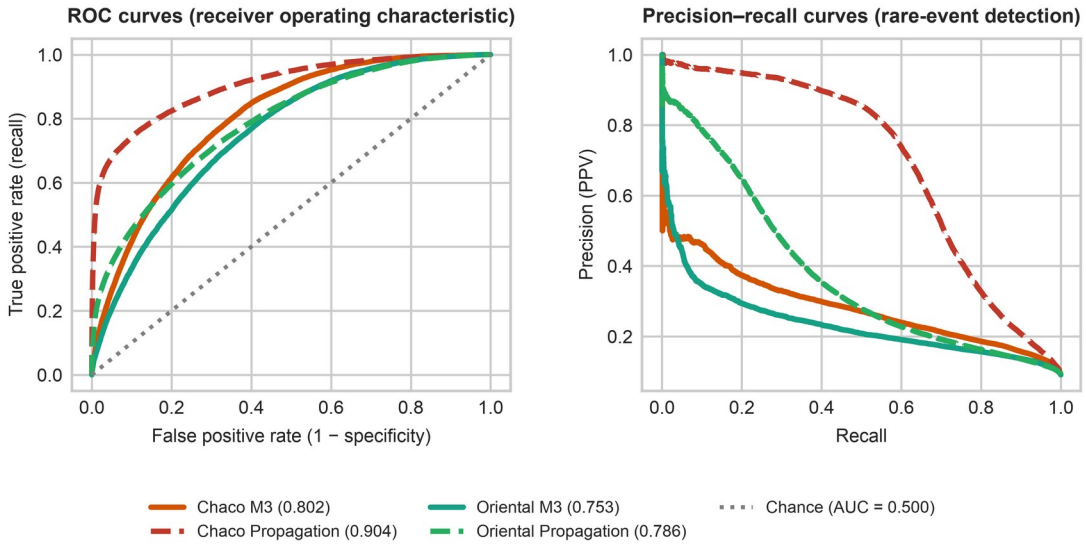


Figure 3: ROC and precision–recall curves for the retrospective D2 test (2024–2025), by region and task. The ignition curves correspond to M3; propagation uses the causal t-1 layer.

3.2 Benchmark against FWI and INPE

Table 3 compares M3 with full FWI and INPE on the same retrospective test. M3 exceeds FWI by ΔROC = +0.2848 in Chaco and +0.1970 in Oriental, and INPE by +0.2633 and +0.2544. All paired comparisons have $p \leq 0.001$ with B = 1 000. The full benchmark, including M2 and propagation rows, is in Supplementary Table S11; the complete significance table is in Supplementary Table S7.

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Table 3: Benchmark on the retrospective 2024–2025 test. ΔROC is M3 minus the reference. FWI and INPE are normalized indices, not calibrated probabilities. Dashes (—) indicate reference models or comparisons not evaluated by construction. The full table is Supplementary Table S11.

Region	Model	ROC-AUC	PR-AUC	ΔROC (M3 − reference)	p
Chaco	M3 ignition	0.8021	0.2842	—	—
Chaco	Full Canadian FWI	0.5173	0.0957	+0.2848	≤0.001
Chaco	INPE Risco de Fogo	0.5389	0.1031	+0.2633	≤0.001
Oriental	M3 ignition	0.7534	0.2340	—	—
Oriental	Full Canadian FWI	0.5564	0.1061	+0.1970	≤0.001
Oriental	INPE Risco de Fogo	0.4990	0.0908	+0.2544	≤0.001
Chaco	Climatological baseline	0.5000	0.0909	+0.3021	—
Oriental	Climatological baseline	0.5000	0.0909	+0.2534	—

Figure 4 shows the corresponding ROC and precision–recall curves. The canonical indices are close to the diagonal in Chaco; in Oriental they are also weak and do not improve materially over a climatological baseline. Full FWI has ROC-AUC 0.5173/0.5564 and INPE 0.5389/0.4990; neither is competitive with M3 on the same retrospective dates. The result is not a claim that the indices are universally invalid, but that their transfer to this subtropical system is not supported. FWI and INPE are normalized danger indices, so their apparent operating point depends strongly on the cutoff chosen on validation. In Chaco, full FWI reaches high recall only at very low precision, while INPE is conservative in both regions; neither behavior translates into a competitive ranking. PR-AUC values are close to the 0.09 case-control baseline, which means that the indices add little discrimination beyond chance on these retrospective dates.

The controlled ablation confirms that socio-ecological variables improve ranking, especially in Oriental. With the same capacity (45/60 rounds and 23/31 leaves), M2 reaches 0.7991/0.7406 and M3 0.8021/0.7534; M1 is the lowest in both regions. The M3 − M2 gain is +0.0030/+0.0128 ROC-AUC and +0.0076/+0.0194 PR-AUC. In Chaco, M3 precision is 32.82 % versus 33.28 % for M2; in Oriental it is 30.13 % versus 26.09 %. The advantage is not uniform and is not presented as a precision gain in both regions. The complete ablation table is Supplementary Table S12. TreeSHAP (Lundberg et al., 2020) ranks historical distance, dry-forest fraction, accessibility, deforestation×VPD pressure and seasonality highest in Chaco; in Oriental,

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historical distance, local focalization, annual agriculture and deforestation pressure dominate. The complete SHAP table and figure are Supplementary Table S13 and Supplementary Figure S1. This does not establish causality, but it shows that the model is not relying on a single weather variable.

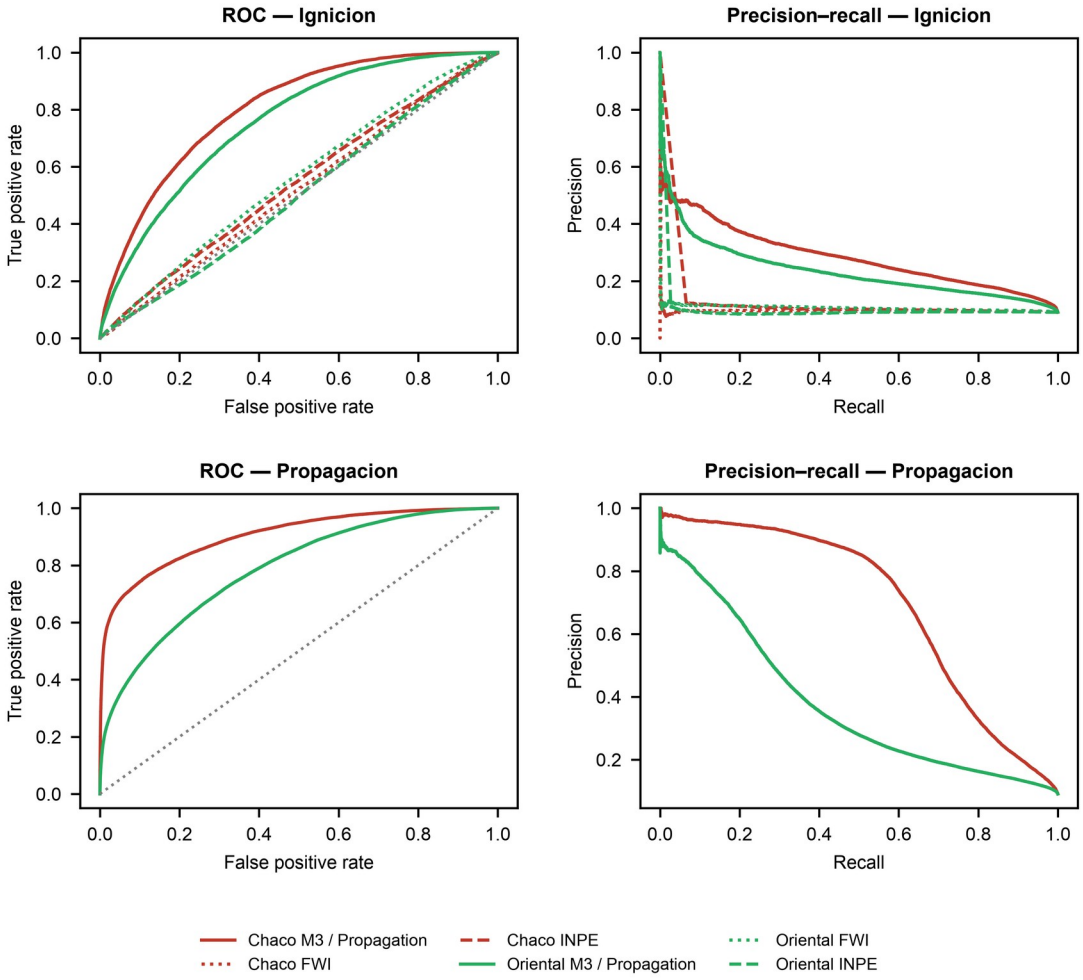


Figure 4: ROC and precision–recall curves for M3 against full Canadian FWI and INPE under D2.

3.3 Temporal and spatial generalization

The M3 advantage is temporal, not spatial. In the 50-km block holdout with a 20-km buffer evaluated in 2023, M2 reaches 0.6773 in Chaco and 0.7470 in Oriental, and M3 reaches 0.6496 and 0.7277; M1 is reported in Supplementary Table S8. M2 is the most robust model in unseen

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blocks. M3 falls from 0.7821 on held-in cells to 0.6496 in the Chaco holdout, and from 0.7473 to 0.7277 in Oriental; M2 falls from 0.7832 to 0.6773 in Chaco but improves from 0.7365 to 0.7470 in Oriental. M1 is more stable, but its absolute level is lower (Supplementary Table S8). Table 4 reports the compact comparison; the full block table and design are in Supplementary Table S8.

Table 4: Development spatial block validation with 50-km blocks and a 20-km buffer, evaluated in 2023. Held-in values are the reference for cells not excluded from training. Full results, including M1, are in Supplementary Table S8.

Region	Model	Held-in ROC-AUC	Holdout ROC-AUC	Holdout PR-AUC
Chaco	M2 base	0.7832	0.6773	0.1760
Chaco	M3 hybrid	0.7821	0.6496	0.1630
Oriental	M2 base	0.7365	0.7470	0.1820
Oriental	M3 hybrid	0.7473	0.7277	0.1608

The development spatial design retains 44 blocks and excludes a 20-km buffer around holdout cells during training; priors are recomputed without held-out fires. An independent confirmation holdout (31 blocks, 68 719 cells), with models retrained without the confirmation blocks and their 20-km buffer, retained M2 as the best model: M2 0.6871/0.6621, M3 0.6757/0.6344 and M1 0.6435/0.6388 in Chaco/Oriental. The gated policy fell back to M2 in the confirmation cells. The result does not support M3 as a spatial extrapolator, and it does not make M1 the best option: M1 was a stable but lower baseline and was rejected as the operational fallback. Full development and confirmation results are in Supplementary Tables S8, S16 and S18; the M3/M2 architecture follows that evidence.

With t-1 weather, M3 retains 0.7831/0.7613 and M2 0.7818/0.7463; M1 is lower and is reported in Supplementary Table S9. The small M3 drop shows that recurrence priors, not same-day weather, dominate the hybrid signal. This is why the current system is a nowcast/hindcast; a real day-ahead system would require forecast weather. Lead-time details are in Supplementary Table S9. As a direct leakage comparator, a coupled model trained on the combined fire sample with t-1 fire inputs did not materially outperform the decoupled task architecture: 0.8862 vs 0.8868 ROC-AUC in Chaco and 0.7871 vs 0.7806 in Oriental. This comparator uses the same gated

case-control sample, so it does not test removal of gating; it only shows that the coupled shortcut does not provide a large advantage on this sample (Supplementary Table S19). A full 3 × 3 D2 sensitivity (15/25/35 km × 7/10/14 d) regenerated the 1:10 sampling and retrained M1, M2 and M3 for every combination. M3 outperformed M2 in all nine combinations on both validation and test, with a positive M3 − M2 ROC-AUC of +0.004 to +0.016, and was the best test model in all nine; its range was 0.771–0.782 in 2023 and 0.785–0.797 in Chaco for 2024–2025, and 0.759–0.768 and 0.750–0.755 in Oriental. The 7-day window gave the highest ROC-AUC in every region × radius pair, and the radius effect was small: 35 km was slightly better in Chaco and 25 km in Oriental. The D2 conclusion is therefore not an artefact of the 25 km/10 d choice (Supplementary Table S20).

3.4 Operational utility

The cost-loss analysis is sobering for ignition: M3 does not clearly outperform M2, and in the Chaco M2 is slightly better at the optimum. The propagation layer does show value near $\alpha = 0.002$. The full cost-loss table and curves are in Supplementary Table S14 and Supplementary Figure S3. The operational product therefore prioritizes propagation and uses ignition as a screening layer.

Cost-loss is defined as the expected value of issuing an alert when the cost of a false alarm is C and the loss avoided by a true alert is L; the ratio $\alpha = C/L$ is varied from 0 to 0.5. This is a deliberately simple decision model, not an estimate of agency budgets. It is useful because it exposes the regime in which a probabilistic layer has value. For ignition, the curves are almost flat near zero; for propagation, the value is concentrated at small α. The result supports the operational choice of prioritizing propagation and treating ignition as a screening layer.

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Conformal coverage is empirical: 90.17 %/90.31 % for a nominal 90 % level and 95.32 %/94.79 % for 95 %. These values are close to nominal but do not constitute a formal guarantee under temporal shift. The result is still useful because it shows that the calibration layer is not grossly overconfident. A reliability diagram and the full conformal table are in Supplementary Table S5. Figure 5 shows an example national alert map. Cells are classified as low, moderate or high; the product is generated from the same frozen test pipeline. The map is illustrative of the output format, not an institutional forecast. It is a communication layer built from the same frozen probabilities and cutoffs; it inherits the calibration and prevalence assumptions of the underlying model. The map should therefore be read together with the population precision in Table 2 and the spatial limitation in Table 4, not as a stand-alone warning product.

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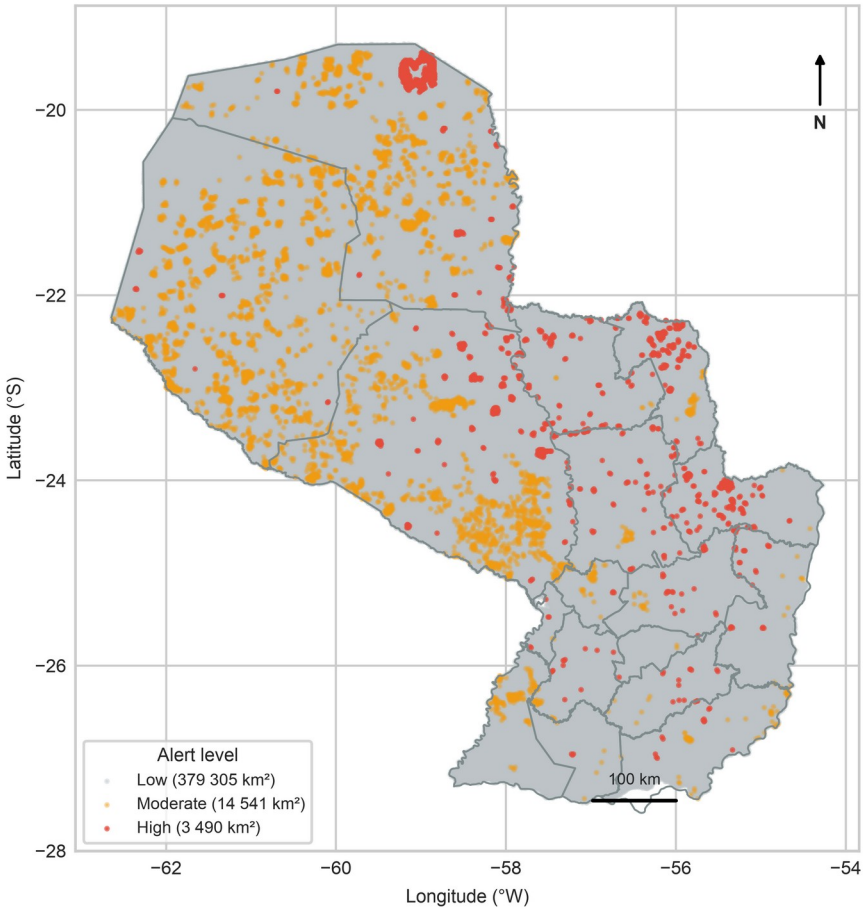


Figure 5: Illustrative national 1-km² alert map for 2024-09-09, showing the spatial distribution of low, moderate and high alert cells.

3.5 Reproducibility checks

The code and data releases are part of the result. The repository contains the target operator, the frozen evaluation scripts, the trained boosters and calibrators, the generated tables and a pinned environment. A clean-environment run of the 76 unit tests covers target construction, VIIRS filtering, climate aggregation, monotonic calibration, metrics and alert routing. The data archive contains the test targets and model probabilities, so the main tables can be regenerated without the raw multi-terabyte archive. The full raw-data pipeline remains available for a complete rebuild.

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Two practical checks are worth reporting. First, the test split is frozen and the cutoff is selected on validation; no test row is used for model selection. Second, the calibration and conformal layers are fitted before the test and never updated. These checks do not eliminate distribution shift, but they remove the most common sources of optimistic bias. The tables in this manuscript are regenerated from the repository and correspond to release v2.0.1; the code and data concept DOIs are reported in the Code and Data Availability sections.

4. Discussion

4.1 Why canonical indices fail

Full FWI and INPE are weak in both regions. The Chaco is moisture-limited and fuel-limited at the same time: fine fuels cure quickly under high VPD, but ignition depends on human access, pasture management and fire history. A single weather index cannot represent that interaction. The Canadian FWI was originally designed for boreal conifer stands, and two of its three fuel-moisture codes (the Duff Moisture Code and the Drought Code) explicitly model the slow drying and rain recharge of loosely and deeply compacted organic layers over days to weeks (Flannigan et al., 2016; Van Wagner, 1987). In the semi-arid Chaco, the load-bearing fuels are fine grassy and shrub fuels that respond within hours to atmospheric demand, while the spatial pattern of ignition is governed by land-management and access-related sources. A purely meteorological index therefore carries no information about where those anthropogenic ignition sources are located, and its ranking skill collapses toward chance under a decoupled D2 evaluation. In the Oriental Region, the signal is more anthropogenic and less dependent on same-day weather. These results do not imply that FWI or INPE are useless globally; they show that their transfer to this subtropical system is not supported by the retrospective test.

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The regional asymmetry is informative. In the Chaco, the dominant land-cover contrast is between dry forest, pastures and natural grasslands; fire is often a management tool in a matrix with high fuel connectivity. In the Oriental Region, the landscape is wetter and more fragmented, and ignition is tightly coupled to roads, farms and the agricultural calendar. The same hybrid architecture therefore operates under different pyrogeographic regimes, and the SHAP rankings reflect that difference: dry-forest fraction is important in the Chaco, while local anthropic concentration is more important in Oriental. A model with a single set of regional weights would blur this contrast.

The weak FWI/INPE performance also has a practical implication: switching to a “better index” is not sufficient. The input representation, the target definition and the evaluation design matter as much as the learner. The benchmark therefore compares the complete computational pipeline, not only the model class.

4.2 What decoupling buys and where it fails

Decoupling produces a useful diagnostic. M3 is a strong temporal nowcast for cells with history, but its advantage disappears in unseen spatial blocks. M2 is the most robust model in those blocks and also outperforms M1 on cells without history, so it replaces M1 as the fallback. This is a double result: the operational layer should not present M3 as a spatial extrapolator, and it should not retain a biophysical fallback that 2023 validation does not support.

The D2 operator is also a guard against leakage. Without it, a model can mix persistence and ignition and obtain impressive but non-operational metrics. The combination of D2, a no-$t-1$ ignition layer and a spatial block holdout is the main methodological contribution. It also explains why a single aggregate score is not enough: the same model can be strong on held-in cells, weak on held-out blocks and weak at population prevalence, and each of these results

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answers a different question. Reporting the three views separately is more informative than selecting the most favorable one.

From a software perspective, the relevant advance is not the size of the model but the auditability of the path from raw observations to a map. A single integer cell identifier, deterministic joins, frozen prevalence metadata and a pinned environment make the pipeline reproducible on a laptop. The comparison with a GeoPandas spatial join is included only to show why the integer lookup matters at national scale; it is not a benchmark of GIS libraries. The open code and data DOIs allow independent re-execution of the main tables and replacement of the model with another learner.

The framework is also intentionally bounded. It does not include a new neural architecture, a data-assimilation scheme or a full fire-spread simulator. Its contribution is to make a difficult applied problem comparable and reproducible. Gradient-boosted trees were retained not as a claim of architectural superiority but because they allow explicit physical monotonicity constraints and full national inference on commodity CPUs (Section 2.4), which is the operational constraint this study targets. That is a narrower claim than a new algorithm, but it is the claim that the evidence supports.

A second lesson is that benchmark design changes the conclusion. If the test had been a random split, M3 would have looked stronger; if the ignition layer had included $t-1$ fire density, it would have looked stronger still. The spatial block and D2 target make those shortcuts visible. This is why the negative result is not a weakness of the paper but part of its contribution: it gives future work a clear baseline to beat without leakage.

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4.3 Implications for computational geoscience

The framework suggests a practical division of labor for geoscientific machine learning. The learner can be replaced; the target operator, the leakage controls and the evaluation harness should not. In this sense, the contribution is a computational protocol and software infrastructure, not a new learner. That distinction matters for computational geoscience because it makes the contribution reusable beyond Paraguay: another team can keep the D2 logic, substitute local fuel and weather layers, and compare its model on the same kind of retrospective test.

The framework also makes failure explicit. A model that is excellent on held-in cells but poor on spatial blocks is not a universal model, and a model that is only competitive on a case-control sample is not an operational system. Reporting both views is more useful than reporting a single high AUC. This is the main methodological lesson of the study, and it generalizes beyond fire: any geoscientific pipeline that combines spatial priors, temporal targets and rare events needs the same separation between ranking skill, calibration and operational prevalence.

4.4 Limitations

Several limitations bound the interpretation of this study:

- **Data and observational constraints.** The study covers one country and two macro-regions without external-country validation. Fuel characterization relies on the annual MapBiomas Collection 2 series (2017–2023) joined by prior year, so 2024 uses the 2023 map with a one-year lag and 2025 extrapolates it with a two-year lag. Active-fire targets depend on VIIRS detections, which can miss sub-canopy or cloud-obscured fires in the absence of an explicit observation model, while ERA5-Land inputs (0.25°, 3-hourly) may smooth local intra-day VPD and wind peaks.

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- **Statistical and generalization limits.** Although D2 sensitivity across 15/25/35-km radii and 7/10/14-day windows confirmed that model rankings are stable under full retraining (Supplementary Table S20), the 2024–2025 test is retrospective rather than blind and shares 90.65 % of its active cells with the training period through temporal overlap. M3 loses its predictive advantage in unseen spatial blocks, which requires M2 as the operational fallback while M1 remains an insufficient baseline; conformal coverage (90.17 %/90.31 % at the 90 % level) is empirical rather than formally guaranteed under temporal shift; and the coupled $t-1$ comparator (Supplementary Table S19) was evaluated on the same gated case-control sample, so it does not test removal of the biophysical gate.
- **Operational and software scope.** Because the pipeline uses observed meteorology up to day *t*, it functions as a nowcast/hindcast rather than a day-ahead forecast supported by numerical weather prediction. Population-level ignition precision is very low (0.082 %/0.198 %), cost-loss shows no clear M3 advantage over M2 for ignition, and the alert layers have not yet been evaluated prospectively with an operational agency. Finally, transferring the pipeline to another country would require rebuilding the 1-km² grid, the historical priors and the prevalence metadata, and the pinned dependency lock records exact package versions without platform-specific wheel hashes (documented in the code repository).

5. Conclusions

We presented a reproducible computational framework for decoupled wildfire ignition and propagation benchmarking in the Gran Chaco and Oriental Region of Paraguay. The D2 target operator separates primary ignition from multi-day persistence, the gated M3/M2 architecture

pairs a recurrence-aware temporal nowcast with a spatially robust fallback, and the evaluation harness makes every comparison auditable. On 2023 validation, M3 reached a ROC-AUC of 0.7844/0.7628 (Chaco/Oriental), whereas M2 proved most robust across both development (0.6773/0.7470) and independent confirmation spatial blocks (0.6871/0.6621; Supplementary Table S18), justifying the replacement of the biophysical baseline M1 as the operational fallback. On the retrospective 2024–2025 test, M3 substantially outperformed the full Canadian FWI (0.8021/0.7534 vs. 0.5173/0.5564) and the INPE Risco de Fogo (0.5389/0.4990), and decoupling did not sacrifice discrimination relative to a coupled $t-1$ baseline (Supplementary Table S19). Nevertheless, population-level ignition precision remains low (0.082 %/0.198 %) and the hybrid advantage is strictly temporal rather than spatial, positioning the system as an operational screening and prioritization tool rather than a stand-alone warning system. Ultimately, the main contribution of this work is not a single susceptibility map but an open, auditable protocol—archived with public code and data—that shows where a geoscientific model holds, where it fails and how to extend the benchmark to other subtropical regions and to day-ahead weather forecasts.

Data availability

Processed data, test predictions, trained models and derived tables are available in the anonymized review repository under CC BY 4.0 and are documented under the data concept DOI provided on the title page. Raw NASA FIRMS, ERA5-Land/CDS, MapBiomas, Hansen GFC and geoBoundaries data are obtained from their original providers under their own terms and are documented with SHA-256 hashes in fuentes/MANIFIESTO_DATOS_v3.csv.

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Competing interests

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Author contributions

Author 1: conceptualization, methodology, software, data curation, formal analysis, validation, visualization, writing—original draft, writing—review and editing, project administration.

Generative AI declaration

Generative AI tools were used for language editing and code assistance. The author reviewed and verified all content and takes full responsibility for the published article.

Computer Code Availability

Name: Incendios_PY

Developer and contact: anonymized for peer review; contact via the Editor-in-Chief

Year first available: 2026

Hardware: x86_64 or ARM64, at least 8 GB RAM

Operating system: macOS 13+, Linux Ubuntu 22.04+, Windows 11 with WSL2

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Language: Python 3.12+; Polars, DuckDB and NumPy for vectorized operations; LightGBM C++ + backend

Dependencies: pinned in requirements-lock.txt

License: MIT

Repository: anonymized review repository (link provided in the submission system)

Code version DOI: provided on the title page

Data version DOI: provided on the title page

Code concept DOI: provided on the title page

Data concept DOI: provided on the title page

User guide: docs/USER_GUIDE.md

Tutorial: docs/TUTORIAL.md

Tests: 76 tests pass with python -m pytest -q tests.

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